ARJUNA JEE 2027

Lecture -03 | Mathematics

BINOMIAL THEOREM

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Topics *to be covered*

A **Number of Rational or Irrational Terms**

B **Finding Coefficients** ★★★★★

C **Question Practice**

D



H.W. Discussion

Example

For 69!, find :

(i) Exponent of 2

(ii) Largest value n such that 69! is divisible by 3^n.

(iii) Exponent of 6 in 69! = 32.

(iv) Number of syphers in end of 69! = exponent of 10 in 69!

$69! = 2^{66} \cdot 3^{32} \cdots$

$= 4^{33} \cdot 3^{32} \cdots$

2 5

exponent of 5 in 69! = 15

(i) $\left[\frac{69}{2}\right] + \left[\frac{69}{2^2}\right] + \left[\frac{69}{2^3}\right] + \cdots = 66.$

(ii) $n = \left[\frac{69}{3}\right] + \left[\frac{69}{3^2}\right] + \left[\frac{69}{3^3}\right] + \left[\frac{69}{3^4}\right] + \cdots = 32.$

(v) exponent of 12 in 69! = 32.

4×3



LAST CLASS RECAP

Answer the following Questions:

1. k^{th} term from end in expansion $(x+y)^n$ is same as k^{th} term from beginning in expansion of $(y+x)^n$. **(True/False)**

2. 13th Term from last in $(2x-3y)^{82}$ is 70th term from starting.

$71^{th} = (82-13+2)^{th}$ from starting **(True/False)**

3. Binomial coefficient of terms equidistant from starting and end are equal. **(True/False)**



LAST CLASS RECAP

Answer the following Questions:

4. **True/False:** $\sum_{r=0}^{7} {}^{7}C_r 2^{7-r} 3^r = 5^7$.

5. **True/False:** $\sum_{r=0}^{10} (-1)^r \cdot {}^{10}C_r 2^{r-1} = 1$.

Note.

$$\sum_{r=0}^{7} {}^{7}C_r \, 2^{7-r} \cdot 3^r = (2+3)^7$$

$$\frac{1}{2} \sum_{r=0}^{10} {}^{10}C_r \, (-2)^r (1)^{10-r} = \frac{(1+(-2))^{10}}{2} = \frac{1}{2}$$

6. Which term of the expansion $\left(x - \frac{1}{x^2}\right)^{32}$ has greatest binomial coefficient middle Term. (T_{17})

6. Value(s) of r for which ${}^{45}C_r$ is maximum.

$$r = \frac{45+1}{2} \text{ OR } \frac{45-1}{2}$$

$$r = 23 \text{ OR } 22.$$

$n = 32$

$\frac{32+1}{2}$, $\frac{32+2}{2}$, $\frac{32+3}{2}$

$\frac{32+2}{2} \rightarrow 17$

BINOMIAL THEOREM

Question [JEE Mains-2024]

85%

If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal

A 6

B 4

C 2

D 9

$$G.T.\ (T_{r+1}) = {}^{10}C_r \left(\sqrt{a}\,x^2\right)^{10-r} \left(\frac{1}{2x^3}\right)^r$$

$$= {}^{10}C_r \frac{(\sqrt{a})^{10-r}}{2^r} \cdot \frac{x^{20-2r}}{x^{3r}}$$

$$105 = {}^{10}C_4 \frac{(\sqrt{a})^6}{2^4}$$

$$x^{20-5r}$$

$$20 - 5r = 0$$

$$4 = r$$

Question [JEE (Adv.)-2023

[Ans. : 3]

Let a and b be two nonzero real numbers. If the coefficient of x^5 in the expansion of $\left(ax^2 + \frac{70}{27bx}\right)^4$ is equal to the coefficient of x^{-5} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^7$, then the value of $2b$ is

(3)

$$T_{r+1} = {}^4C_r (ax^2)^{4-r} \left(\frac{70}{27bx}\right)^r$$

$$= {}^4C_r \cdot \frac{a^{4-r} \cdot 70^r}{(27b)^r} \cdot x^{8-3r}$$

$8-3r = 5$

$r = 1$

$$T_{r+1} = {}^7C_r (ax)^{7-r} \left(\frac{-1}{bx^2}\right)^r$$

$$= {}^7C_r \cdot \frac{a^{7-r} \cdot (-1)^r}{b^r} \cdot x^{7-3r}$$

$7-3r = -5$

$12 = 3r$

$r = 4$

coeff of $x^5 = {}^4C_1 \cdot \frac{a^3 \cdot 70}{27b} = {}^7C_4 \frac{a^3 \cdot (-1)^4}{b^4}$

$b = \frac{3}{2}$

Question [JEE Mains-2024]

Let m and n be the coefficients of seventh and thirteenth terms respectively in the expansion of $\left(\frac{1}{3}x^{\frac{1}{3}} + \frac{1}{2x^{\frac{2}{3}}}\right)^{18}$. Then $\left(\frac{n}{m}\right)^{\frac{1}{3}}$ is :

76%

A $\frac{4}{9}$

B $\frac{1}{9}$

C $\frac{1}{4}$

D $\frac{9}{4}$

$$T_{r+1} = {}^{18}C_r \cdot \left(\frac{1}{3}x^{\frac{1}{3}}\right)^{18-r} \cdot \left(\frac{1}{2x^{\frac{2}{3}}}\right)^{r}$$

$$T_{r+1} = {}^{18}C_r \cdot \left(\frac{1}{3}\right)^{18-r} \cdot \left(\frac{1}{2}\right)^{r} \cdot x^{6-r}$$

$r=6$: $T_7 = {}^{18}C_6 \left(\frac{1}{3}\right)^{12} \cdot \left(\frac{1}{2}\right)^{6}$ → m

$r=12$: $T_{13} = {}^{18}C_{12} \left(\frac{1}{3}\right)^{6} \cdot \left(\frac{1}{2}\right)^{12} x^{-6}$ → n

$$\frac{n}{m} = \frac{{}^{18}C_{12}\left(\frac{1}{3}\right)^{6}\left(\frac{1}{2}\right)^{12}}{{}^{18}C_{6}\left(\frac{1}{3}\right)^{12}\left(\frac{1}{2}\right)^{6}} = \left(\frac{\frac{1}{2}}{\frac{1}{3}}\right)^{6}$$

$$\frac{n}{m} = \left(\frac{3}{2}\right)^{6}$$

$$\left(\frac{n}{m}\right)^{\frac{1}{3}} = \left(\left(\frac{3}{2}\right)^{6}\right)^{\frac{1}{3}} = \left(\frac{3}{2}\right)^{2}$$

RAPID FIRE DIBY'S

DIBY-01

Find the term independent of x in the expansion of $\left(\frac{3x^2}{2} - \frac{1}{3x}\right)^9$.

(A) $\frac{7}{54}$ (B) $-\frac{7}{18}$ (C) $\frac{7}{18}$ (D) $-\frac{7}{54}$

[Ans. : (C)]

DIBY-02

Find n if ratio of the 7th term from the beginning to the 7th term from the end in the expansion of $\left(2^{\frac{1}{3}} + 3^{-\frac{1}{3}}\right)^n$ is $\frac{1}{6}$.

(A) 9 (B) 8 (C) 7 (D) 6

[Ans. : (A)]

RAPID FIRE DIBY'S

DIBY-03

Find the coefficient of x^{32} in the expansion of $\left(x^4 - \frac{1}{x^3}\right)^{15}$

(A) −1365 (B) 1365 (C) −3003 (D) 3003

[Ans. : (B)]

DIBY-04

The 4th term from the end in the expansion of $\left(\frac{3}{x^2} - \frac{x^3}{6}\right)^7$ is

(A) $\frac{35}{8}x^3$ (B) $-\frac{35}{48}x^3$ (C) $-\frac{35}{8}x^6$ (D) $\frac{35}{48}x^6$

[Ans. : (D)]

DIBY-05

Find the greatest value of the term independent of x in the expansion of $\left(x\sin\alpha + \frac{1}{x}\cos\alpha\right)^{10}$, where $\alpha \in \mathbb{R}$.

(A) $\frac{63}{8}$ (B) $\frac{63}{16}$ (C) $\frac{63}{32}$ (D) $\frac{63}{64}$

[Ans. : (A)]

FINDING RATIONAL & IRRTATIONAL TERMS

Question [JEE Mains-2016]

The total number of irrational terms in the binomial expansion $(7^{1/5} - 3^{1/10})^{60}$ is

A 48

B 49

C 54

D 55

$\left(7^{\frac{1}{5}} - 3^{\frac{1}{10}}\right)^{60}$ → Total terms = 60 + 1 = 61.

$T_{r+1} = {}^{60}C_r \cdot \left(7^{\frac{1}{5}}\right)^{60-r} \cdot \left(-3^{\frac{1}{10}}\right)^{r}$

$= {}^{60}C_r \cdot (7)^{\frac{60-r}{5}} \cdot (-1)^{r} \cdot 3^{\frac{r}{10}}$

$\left(7^{\frac{1}{5}} - 3^{\frac{1}{10}}\right)^{60}$ → Total terms = 60 + 1 = 61.

$T_{r+1} = {}^{60}C_r \cdot \left(7^{\frac{1}{5}}\right)^{60-r} \cdot \left(-3^{\frac{1}{10}}\right)^{r}$

$T_{r+1} = {}^{60}C_r \cdot (7)^{\frac{60-r}{5}} \cdot (-1)^{r} \cdot 3^{\frac{r}{10}}$

${}^{60}C_r$ → $r \in W$, $0 \le r \le 60$

Rational Terms = 7

IrrRational Terms = 61 − 7 = 54

Rational ⇒ $\frac{60-r}{5}$ & $\frac{r}{10}$ → integer. (both)

$\frac{60-r}{5} = 12 - \frac{r}{5} = \text{Int}$

$r = 0, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60$

$\frac{r}{10} = \text{Int}$

$r = 0, 10, 20, 30, 40, 50, 60$

$r = 0, 10, 20, 30 \ldots 60$

$r = 1$ → ${}^{60}C_1 \cdot 7^{\frac{59}{5}} \cdot 3^{\frac{1}{10}}$ → irrational

Question [Multiple Correct]

In the expansion $\left(4^{\frac{1}{3}}+\frac{1}{6^{1/4}}\right)^{20}$ which of the following hold good?

A no. of irrational term = 19

B middle term is irrational

C no. of rational term = 2

D none

$21-2$

$T_{r+1} = {}^{20}C_r \left(4^{\frac{1}{3}}\right)^{20-r} \left(\frac{1}{6^{\frac{1}{4}}}\right)^{r}$

$= {}^{20}C_r \, 2^{\frac{40-2r}{3}} \, \frac{1}{2^{\frac{r}{4}} 3^{\frac{r}{4}}}$

$\frac{20+1}{2}$, $\frac{20+2}{2}$, $\frac{20+3}{2}$

$\times$, 11th , $\times$

$T_{r+1} = {}^{20}C_r \, 2^{\frac{160-11r}{12}} \, 3^{\frac{-r}{4}}$

$$T_{r+1} = {}^{20}C_r \, 2^{\frac{160-11r}{12}} \cdot 3^{\frac{-r}{4}}$$

$r \in W$

$0 \le r \le 20$

$\frac{160-11r}{12}$ & $\frac{-r}{4}$ $\rightarrow$ Integer.

$\frac{160-11r}{12} \rightarrow$ Integer.

$13 + \frac{4-11r}{12}$

$\frac{-r}{4} \rightarrow$ Integer.

'r' is multiple of 4.

$r = 12 \Rightarrow 13 + \frac{4-132}{12}$ ×

$r = 16 \Rightarrow 13 + \frac{4}{12} - \frac{11\times16}{12}$ ×

$r = 20 \Rightarrow 13 + \frac{4}{12} - \frac{11\times20}{12}$

$13 - 18$ ✓

$r = 0 \Rightarrow 13 + \frac{4}{12}$ ×

$r = 4 \Rightarrow 13 + \frac{4-44}{12}$ ×

$r = 8 \Rightarrow 13 + \frac{4-88}{12}$ ✓

$r = 0, 4, 8, 12, 16, 20$

$r = 8, 20$

Rational = 2.

Question [JEE Mains-2020]

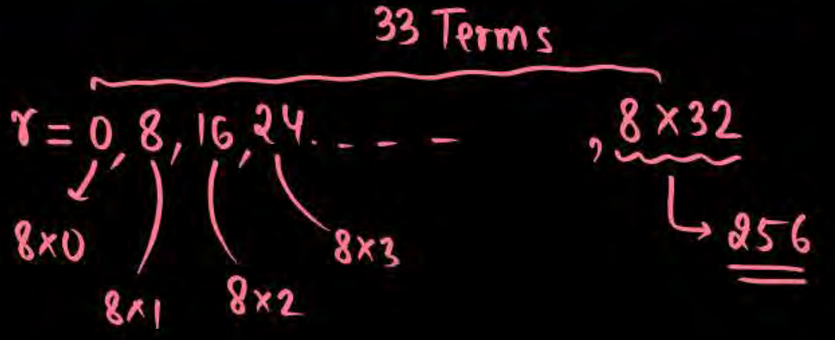
If the number of integral terms in the expansion of $\left(3^{1/2}+5^{1/8}\right)^{n}$ is exactly 33, then the least value of n is

A 264

B 128

C 256

D 248



$${}^{n}C_{r}\left(3^{\frac{1}{2}}\right)^{n-r}\left(5^{\frac{1}{8}}\right)^{r}$$

$$T_{r+1} = {}^{n}C_{r}\cdot 3^{\frac{n-r}{2}}\,5^{\frac{r}{8}}$$

$0 \le r \le n$

$n = 256, 258, 260, 262, (264)$

$n \to$ even

$\frac{n-r}{2} = \text{Int.}$ & $\frac{r}{8} = \text{Int.}$

$\frac{n}{2} - \frac{r}{2} = \text{Int.}$

$r = 0, 8, 16, 24, 32, 40 \ldots . 256$

Question [JEE Mains-2024]

Number of integral terms in the expansion of $\left\{7^{\left(\frac{1}{2}\right)}+11^{\left(\frac{1}{6}\right)}\right\}^{824}$ is equal to _ _ _ _ _ _ _ _

Ans: 138

QIBY — A

RAPID FIRE ROUND

1. Coefficient of x^r in $(1+x)^n$ is ${}^nC_r$.

2. Coefficient of x^9 in $(1+3x+3x^2+x^3)^{15}$ is ${}^{45}C_9$.

3. Coefficient of x^{10} in $x^3(2-3x)^9$ is $-{}^9C_7 \cdot 2^2 \cdot 3^7$.

4. Coefficient of x^{10} in $\frac{(2x+5)^{17}}{x^4}$ is coeff. of x^{14} in $(2x+5)^{17}$ $= {}^{17}C_3 \cdot 2^{14} \cdot 5^3$

$(2x+5)^{17} = {}^{17}C_r (2x)^{17-r} \cdot 5^r = {}^{17}C_r \, 2^{17-r} \cdot 5^r \, x^{17-r}$

$((1+x)^3)^{15} = (1+x)^{45}$

$= {}^{45}C_r \, x^r$

coeff of x^{10} in $x^3(2-3x)^9$ = coeff of x^7 in $(2-3x)^9$

$(2-3x)^9 = {}^9C_r \, 2^{9-r} \cdot (-3)^r \cdot x^r$

Blue बादल (KKPNCC)

Before finding coefficient

Always simplify → Given expression

Acidic Question

If $(1 + x + x^2 + x^3)^{15} = a_0 + a_1 x + a_2 x^2 + \ldots.$, then find a_5 & also find total number of terms.

a_5 → coeff. of x^5

$$\left(1+x+x^2(1+x)\right)^{15} = \left((1+x)(1+x^2)\right)^{15}$$

coeff. of $x^5 \rightarrow (1+x)^{15} \cdot (1+x^2)^{15}$

Q iBY — (B)

a_7 = coeff. of x^7

$^{15}C_r\, x^{(r)}$

coeff. of $x^5 \rightarrow (1+x)^{15}\,(1+x^2)^{15}$

$^{15}C_5\, x^5 \times {}^{15}C_0\, x^0 = {}^{15}C_5 \times 1\, x^5$

$^{15}C_3\, x^3 \times {}^{15}C_1\, x^2 = {}^{15}C_3\, {}^{15}C_1\, x^5$

$^{15}C_1\, x^1 \times {}^{15}C_2\, x^4 = {}^{15}C_1\, {}^{15}C_2\, x^5$

$$\left({}^{15}C_5 + {}^{15}C_3\, {}^{15}C_1 + {}^{15}C_1\, {}^{15}C_2\right) x^5$$

fnf

$(1+x^2)^{15}$

$^{15}C_r\, (x^2)^r$

$^{15}C_0 + {}^{15}C_1 x^2 + {}^{15}C_2 x^4 + {}^{15}C_3 x^6 \ldots\ldots$

Acidic Question

If $(1 + x + x^2 + x^3)^{15} = a_0 + a_1 x + a_2 x^2 + \ldots.$, then find a_5 & also find total number of terms.

$$\left(1+x+x^2(1+x)\right)^{15} = (1+x)^{15} \cdot (1+x^2)^{15}$$

$$(1+x+x^2+x^3)^{15} = a_0 + a_1 x + a_2 x^2 + \ldots\ldots\ldots + a_{45} x^{45}$$

T.T. = 46

N.O.T. = (max. power + 1)
(provided all terms should be present)

(A) 16^2

(B) 32

(C) 46

(D) 52.

$$(1+x+x^2+x^3)^{15} = a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots + a_{45}x^{45}$$

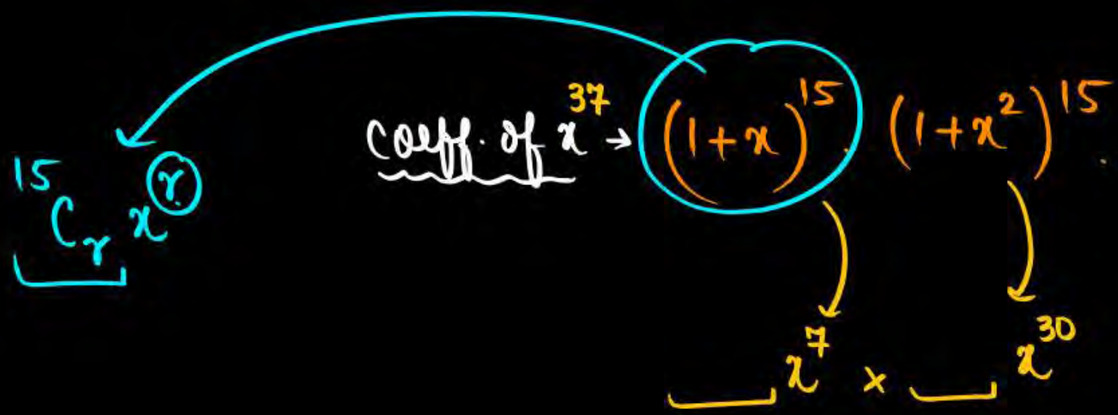
T.T. = 46

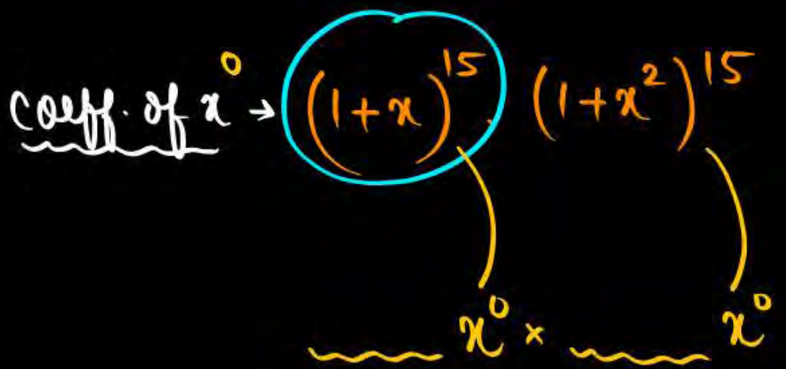
$x \to x^2$

$$(1+x^2+x^4+x^6)^{15} = a_0 + a_1x^2 + a_2x^4 + \ldots\ldots + a_{45}x^{90}$$

T.T. = 91 ✗

T.T. = 46 ✓


coeff. of x^{37} → $(1+x)^{15}$ $(1+x^2)^{15}$
$^{15}C_r\, x^{(r)}$
$x^7 \times x^{30}$


coeff. of x^0 → $(1+x)^{15}$ $(1+x^2)^{15}$
$x^0 \times x^0$

Question [JEE Mains-2022]

The term independent of x in the expression of

$$\left(1 - x^2 + 3x^3\right)\left(\frac{5}{2}x^3 - \frac{1}{5x^2}\right)^{11}, \quad x \neq 0 \text{ is}$$

A $\frac{7}{40}$

B $\frac{33}{200}$

C $\frac{39}{200}$

D $\frac{11}{50}$

coeff of x^0

$1\,x^0 \times \quad 0 \quad x^0 \quad = \quad 0\,x^0$

$-1\,x^2 \times {}^{11}C_7\left(\frac{5}{2}\right)^4\left(\frac{-1}{5}\right)^7 x^{-2} = +{}^{11}C_7\left(\frac{5}{2}\right)^4 \cdot \frac{1}{5^7}\,x^0$

$3x^3 \times \quad 0 \quad x^{-3} \quad = \quad 0\,x^0$

$$\frac{2}{10} \times 3 \times 11 \times \frac{1}{125} = \frac{33}{200}$$

$$T_{r+1} = {}^{11}C_r\left(\frac{5}{2}x^3\right)^{11-r}\left(\frac{-1}{5x^2}\right)^r$$

$$= {}^{11}C_r\left(\frac{5}{2}\right)^{11-r}\left(\frac{-1}{5}\right)^r x^{33-5r}$$

$r \in W$

$0 \le r \le 11$

$x^0, x^2, x^4, x^5, x^8 \ldots$ $\qquad$ $x^0, x^3, x^5, x^9, x^{12} \ldots$

\# coeff of $x^5 \rightarrow (1+x^2)^7 \cdot (2+x^3)^5$

$^7C_1\, x^2 \times {}^5C_1 \cdot 2^4 x^3 = (35 \times 16)x^5$ ✓

$(1+x^2)^7$

$\rightarrow {}^7C_r (x^2)^r$

$(2+x^3)^5$

$\rightarrow {}^5C_r \cdot 2^{5-r} \cdot (x^3)^r$

Question [JEE (Adv.)-2014]

Coefficient of x^{11} in the expansion of $(1+x^2)^4(1+x^3)^7(1+x^4)^{12}$ is



A 1051

B 1106

C 1113

D 1120

Question [JEE Mains-2024]

If the constant term is expansion of $\left(1+2x-3x^3\right)\left(\frac{3}{2}x^2-\frac{1}{3x}\right)^9$ is p, then $108p$ is equal to



DO IT BY YOURSELF (DIBY)

DIBY-01

The coefficient of x^{-6} in the expansion of $\left(\frac{4x}{5}+\frac{5}{2x^2}\right)^9$, is ____ [JEE Mains-2023]

[Ans. : 5040]

$r=21 \rightarrow m=43$ (x, x) $\quad r=19 \rightarrow m=13$

$\# m = \frac{2r+1}{22-r}$, $r \in$ whl, $0 \le r \le 22$

DIBY-02

The natural number m, for which the coefficient of x^1 in the binomial expansion of $\left(x^m+\frac{1}{x^2}\right)^{22}$ is 1540, is ____ [JEE Mains-2020]

[Ans. : 13]

$^{22}C_r (x^m)^{22-r} \left(\frac{1}{x^2}\right)^r = {}^{22}C_r \cdot x^{22m-rm-2r}$

$22m-rm-2r=1$

$^{22}C_r = 1540$

DIBY-03

The sum of the real values of x for which the middle term in the binomial expansion of $\left(\frac{x^3}{3}+\frac{3}{x}\right)^8$ equals 5670 is [JEE Mains-2019]

(A) 0 (B) 8 (C) 6 (D) 4

[Ans. : (A)]

$\frac{8+1}{2}, \frac{8+2}{2}, \frac{8+3}{2}$ → 5

$T_5 = {}^8C_4 \frac{x^{12-8}}{3^4} \times \frac{3^4}{x^4} = 5670$

$x^8 = \frac{5670}{{}^8C_4}$

$^{20}C_{10}\,x^{10}$ = M.T. = T_{10+1}

DIBY-04

The ratio of the coefficient of the middle term in the expansion of $(1+x)^{20}$ and the sum of the coefficients of two middle terms in the expansion of $(1+x)^{19}$ is

[JEE Mains-2021]

$\text{Inf} = \dfrac{^{20}C_{10}}{^{19}C_9 + {}^{19}C_{10}}$ ✓

$T_{10} = {}^{19}C_9\,x^9$

$T_{11} = {}^{19}C_{10}\,x^{10}$ · T_{10}, T_{11}

[Ans. : 1]

DIBY-05

If the constant term, in binomial expansion of $\left(2x^r + \frac{1}{x^2}\right)^{10}$ is 180, then r is

[JEE Mains-2021]

$^{10}C_k\,2^{10-k}\cdot x^{(10r-rk-2k)} = {}^{10}C_k\,(2x^r)^{10-k}\cdot\left(\frac{1}{x^2}\right)^k$ · $10r-rk-2k=0$

[Ans. : 8]

DIBY-06

If the term independent of x in the expansion of $\left(\sqrt{a}x^2 + \frac{1}{2x^3}\right)^{10}$ is 105, then a^2 is equal to

[JEE Mains-2024]

(A) 2 (B) 4 (C) 6 (D) 9

[Ans. : (B)]

DIBY-07

If the constant term in the expansion of $\left(\frac{\sqrt[5]{3}}{x}+\frac{2x}{\sqrt[3]{5}}\right)^{12}, x \neq 0$ is $\alpha \times 2^8 \times \sqrt[5]{3}$, then 25α is equal to

[JEE Mains-2024]

[Ans. : 693]

DIBY-08

If the term without x in the expansion of $\left(x^{2/3}+\frac{\alpha}{x^3}\right)^{22}$ is 7315, then $|\alpha|$ is equal to

[JEE Mains-2023]

[Ans. : 1]

DIBY-09

If the coefficient of x^4, x^5 and x^6 in the expansion of $(1+x)^n$ are the arithmetic progression then the maximum value of n is

[JEE Mains-2024]

(A) 7 (B) 21 (C) 28 (D) 14

[Ans. : (D)]

DIBY-10

Let the coefficient of three consecutive terms in the binomial expansion of $(1 + 2x)^n$ be in the ratio **2 : 5 : 8**. Then the coefficient of the term, which is in the middle of these three terms, is

[JEE Mains-2023]

[Ans. : 1120]

DIBY-11

If the coefficient of x^7 in $\left(x^2 + \frac{1}{bx}\right)^{11}$ and x^{-7} in $\left(x - \frac{1}{bx^2}\right)^{11}$ $(b \neq 0)$, are equal, then the value of b is equal to

[JEE Mains-2021]

(A) 2 (B) –1 (C) 1 (D) –2

[Ans. : (C)]

DIBY-12

If the co-efficient of x^9 in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ and the co-efficient x^{-9} in $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$ are equal to, then $(\alpha\beta)^2$ is equal to

[JEE Mains-2023]

[Ans. : 1]

It is never about end result

It's always about Journey

Thank You

#futureIITians